

第六章 电磁场

$$d\vec{F} = I d\vec{l} \times \vec{B}$$

$$\int q \vec{v} \times \vec{B}$$

为什么导体内载流子不会飞出?

产生了反向的静电场

$$\text{电子: } j \text{ 与 } j' = -eE \text{ 平衡 } j = j'$$

$$\text{静电: } \oint \vec{E} \cdot d\vec{l} = 0 \quad \Rightarrow \quad j = j'$$

宏观

安培力

$$\vec{F} = q(\vec{v} \times \vec{B} + \vec{E})$$

$$\vec{F} \leftarrow \left[\begin{array}{c} \vec{v} \\ \vec{E} + \vec{v} \times \vec{B} \end{array} \right] q$$

$$\text{例: } \vec{E} = (E_0 \sin(\omega t), 0, 0)$$

$$\vec{B} = (0, B_0 \cos(\omega t), 0)$$

$$\vec{v}|_0 = (0, 0, 0)$$

① $\omega_1 = \omega_2 = \omega$ ② $\omega_1 \neq \omega_2$ 求 q 的空间轨迹

$$m\ddot{\vec{r}} = q(\dot{\vec{r}} \times \vec{B} + \vec{E})$$

$$m\ddot{x} = q[-\frac{1}{2}B_0 \cos(\omega t) + E_0 \sin(\omega t)] \quad ①$$

$$m\ddot{y} = 0 \quad ②$$

$$m\ddot{z} = q \times B_0 \cos(\omega t) \quad ③$$

$$\text{③} \rightarrow \dot{x} = \frac{m}{qB_0} \ddot{z} \cos(\omega t)$$

$$\ddot{x} = \frac{m}{qB_0} [\ddot{z} \cos(\omega t) - \omega \dot{z} \sin(\omega t)]$$

代入 ①

$$\frac{m^2}{qB_0} [\ddot{z} \cos(\omega t) - \omega \dot{z} \sin(\omega t)] = q[-\frac{1}{2}B_0 \cos(\omega t) + E_0 \sin(\omega t)]$$

$$\text{记 } k = \frac{qB_0}{m}$$

上式化为二阶线性非齐次方程

$$\ddot{z} + \omega \sin(\omega t) \dot{z} + k^2 \cos^2(\omega t) z = k^2 E_0 \sin(\omega t) \cos(\omega t)$$

对应的齐次方程为

$$\ddot{z} + \omega \sin(\omega t) \dot{z} + k^2 \cos^2(\omega t) z = 0$$

代入试探解 $z = A \sin(\Omega t)$

$$\text{得 } -\Omega^2 \sin(\Omega t) + \omega \sin(\omega t) (\cos(\Omega t)) + k^2 \cos^2(\omega t) \sin(\Omega t) = 0$$

$$-\Omega^2 \sin(\Omega t) + \frac{1}{2} \omega [\sin(\Omega + \omega)t + \sin(\Omega - \omega)t]$$

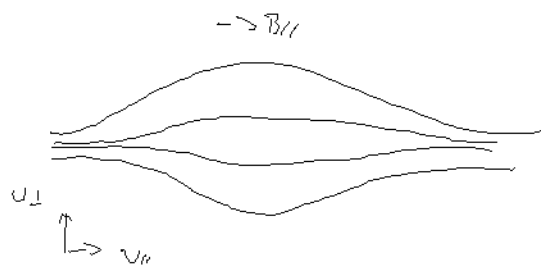
$$+ \frac{1}{4} [k^2 \sin(\Omega t) + \frac{1}{4} k^2 \sin(\Omega + 2\omega)t + \sin(\Omega - 2\omega)t]$$

磁约束:

粒子运动中磁矩守恒.

$$J_c = q v_z B_0 = m \frac{v_\perp^2}{r}$$

$$J_0 = -q v$$



等离子体振荡

电子数密度微扰:

$$n_e(\vec{r}, t) = n_0 + n_1(\vec{r}, t)$$

连续性方程:

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e \vec{v}) = 0$$

$$\nabla \cdot (n_0 \vec{v}) = \nabla \cdot (n_0 \vec{v}) + \nabla \cdot (n_1 \vec{v})$$

$$\hookrightarrow \frac{\partial n_1}{\partial t} + n_0 \nabla \cdot \vec{v} = 0$$

$$\left\{ \begin{array}{l} m \frac{\partial \vec{v}}{\partial t} = -e \vec{E} \quad \text{①} \end{array} \right.$$

$$\nabla \cdot \vec{E} = -\frac{e}{\epsilon_0} [n_e - n_0] = -\frac{en_1}{\epsilon_0} \quad \text{②}$$

$\nabla \cdot \text{①}$

$$\hookrightarrow \frac{\partial}{\partial t} (\nabla \cdot \vec{v}) = -\frac{e}{m} (\nabla \cdot \vec{E}) = -\frac{e^2}{m\epsilon_0} n_1$$

$$= -\frac{1}{n_0} \frac{\partial}{\partial t} \left(\frac{\partial n_1}{\partial t} \right)$$

$$\Rightarrow \frac{\partial^2 n_1}{\partial t^2} - \frac{n_0 e^2}{m\epsilon_0} n_1 = 0 \quad \omega = \sqrt{\frac{n_0 e^2}{m\epsilon_0}}$$

截止频率:

$$\nabla \cdot \vec{E} = 0 \quad (\text{在等离子体中}) \quad \nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{B} = \mu_0 (\vec{j} + \frac{\partial \vec{D}}{\partial t})$$

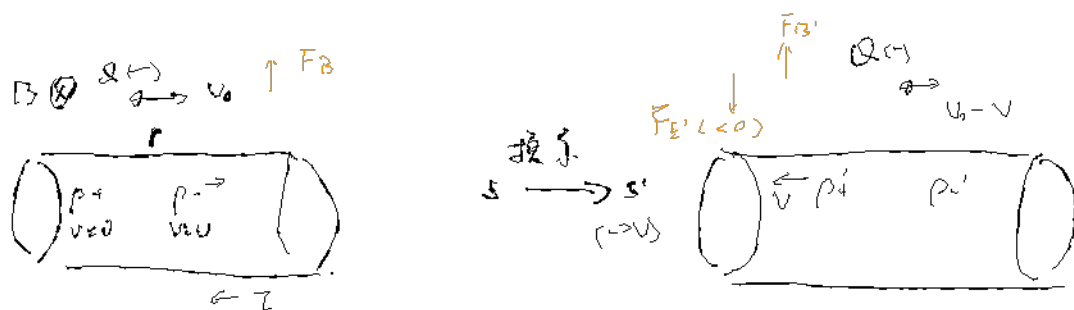
罗伦兹实验 $\rightarrow$ 运动电荷产生磁场

运动电荷 $\rightarrow \vec{B} = \frac{\mu_0}{4\pi} \frac{Q \vec{v} \times \vec{r}}{r^3} \quad \vec{E} = \frac{Q \vec{r}}{4\pi \epsilon_0 r^3}$

$\Rightarrow \vec{B} = \mu_0 \epsilon_0 \vec{v} \times \vec{E} = \frac{1}{c} \vec{v} \times \vec{E}$

相对论效应:

点电荷 Q 在电流为 I 的长直导线附近以 v_0 运动



$F_B = \frac{\mu_0}{2\pi r} \cdot Q v_0$

$= \frac{\mu_0 \rho_- s Q}{2\pi r} v_{0U}$

$I = \rho_- s U$

$S \rightarrow S'$:

对于在 S 系静止的电荷 Δ

S' 中 $L = \gamma L_0 \quad \rho_+' = \frac{Q L_0}{L} = \frac{1}{\gamma} \rho_+$

对于在 S' 系中静止的电荷 Δ .

S 中 $L_0 = \gamma L \quad \rho_- = \frac{Q L}{L_0} = \gamma \rho_-$

在 S' 系导线中: $\beta = \frac{v}{c} \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}$

$\rho' = \rho_+' + \rho_- = \left[\frac{1}{\gamma} - \gamma \right] \rho_0 = -\frac{\beta^2}{1-\beta^2} \rho_0$

$E' = \frac{1}{2\pi r L} \cdot \frac{\rho'_+ s L}{\epsilon_0}$

$= -\frac{\beta^2}{1-\beta^2} \frac{\rho_0 s}{2\pi \epsilon_0 r}$

$$F_z' = - \frac{\rho^2}{(1-\beta^2)} \cdot \frac{\rho_0 s Q}{2\lambda r} \quad \frac{1}{\gamma_0 \mu_0} = c^2 = - \sigma \frac{\rho_0 s Q}{2\lambda r} v^2$$

$$F_B' = \mu_0 \rho_+ \frac{V}{2\lambda r} s Q (V_0 - V) \\ = \frac{1}{\sigma} \mu_0 \rho_0 \frac{s}{2\lambda r} Q (V_0 - V) v$$

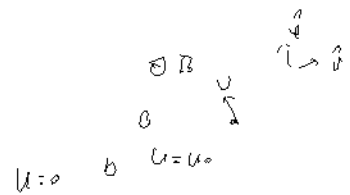
$$\Rightarrow \text{在 } S' \text{ 系中 } Q \text{ 受合力: } F_B' - F_z' = \mu_0 \rho_0 \frac{s Q}{2\lambda r} \left(\frac{1}{\sigma} (V_0 - V) v + \sigma v^2 \right)$$

$$\text{在 } S \text{ 系中 } Q \text{ 受合力: } F_B = \mu_0 \rho_0 \frac{s Q}{2\lambda r} V_0 v$$

$$\text{则! 证, } F_B' + F_z' = F_B$$

05-10

例 1: 研究电子在均匀场中运动



$$E(r) \cdot 2\pi r l = \frac{Q_0}{\epsilon_0} = C$$

$$\rightarrow E(r) = \frac{C}{2\pi r l}$$

$$\int_a^b E(r) dr = \frac{C}{2\pi l} \ln \frac{b}{a} = U_0$$

$$\rightarrow C = 2\pi l \frac{U_0}{\ln \frac{b}{a}} \quad E(r) = \frac{U_0}{r \ln \frac{b}{a}}$$

$$\begin{aligned} \vec{F}_L &= -e \vec{v} \times \vec{B} = e (\dot{r} \hat{r} + r \dot{\varphi} \hat{\varphi}) \times \vec{B} \\ &= -e (r \dot{\varphi} B \hat{r} - \dot{r} B \hat{\varphi}) \end{aligned}$$

$$\text{运动方程} \begin{cases} m(\ddot{r} - r\dot{\varphi}^2) = -eE(r) - e r \dot{\varphi} B = -e \left(\frac{U_0}{\ln \frac{b}{a}} + r \dot{\varphi} B \right) & (1) \\ m(2r\dot{\varphi} + r\ddot{\varphi}) = e \dot{r} B & (2) \end{cases}$$

$$(2) \rightarrow m \frac{d(r^2 \dot{\varphi})}{dt} \cdot \frac{1}{r} = e \frac{dr}{dt} B$$

$$d \left(m r^2 \dot{\varphi} - \frac{1}{2} e B r^2 \right) = 0$$

$$\text{守恒运动量} \quad L = m r^2 \dot{\varphi} - \frac{1}{2} e B r^2 = \text{const.}$$

例 6.1-16

$$\begin{cases} m\ddot{x} = qB\dot{y} & (1) \\ m\ddot{y} = -qB\dot{x} & (2) \\ m\ddot{z} = 0 & (3) \end{cases}$$

$$(3) \rightarrow z = v_0 \cos \theta t$$

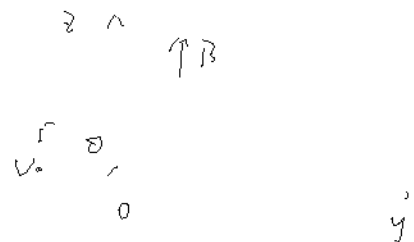
$$(2) \rightarrow \ddot{x} = -\frac{m}{qB} \ddot{y} \quad \text{代(1)} \quad (4)$$

$$\ddot{y} = -\left(\frac{qB}{m}\right)^2 y \rightarrow y = A_y \cos\left(\frac{qB}{m}t\right) + B_y \sin\left(\frac{qB}{m}t\right)$$

$$\text{代(1)} \quad \dot{y}(0) = 0, \quad \ddot{y}(0) = -\frac{qB}{m} v_0 \sin \theta$$

$$\rightarrow A_y = 0, \quad B_y = -v_0 \sin \theta$$

$$\dot{y} = -v_0 \sin \theta \sin\left(\frac{qB}{m}t\right)$$



$$\vec{B} \text{ 沿 } z \text{ 轴: } \ddot{x} = -\left(\frac{qB}{m}\right)^2 x \quad \ddot{y} = \frac{qB}{m} \dot{z} = \frac{qB}{m} v_0 \sin \theta \cos \left(\frac{qB}{m} t\right) \\ = v_0 \sin \theta \cos \left(\frac{qB}{m} t\right)$$

$$\text{在 } t=0, x=y=z=0$$

$$\rightarrow x = \frac{m}{qB} v_0 \sin \theta \sin \left(\frac{qB}{m} t\right)$$

$$y = \frac{m}{qB} v_0 \sin \theta \left(\cos \frac{qB}{m} t - 1 \right)$$

$$z = v_0 \cos \theta t$$

$$\hat{x} = \hat{B} \times (\hat{v}_0 \times \hat{B}) = v_0 (\hat{B} \cdot \hat{B}) - \hat{B} (\hat{v}_0 \cdot \hat{B}) = \hat{v}_0 - (\hat{v}_0 \cdot \hat{B}) \hat{B}$$

$$\hat{y} = \hat{v}_0 \times \hat{B}$$

$$\hat{z} = \hat{B}$$

$$\rightarrow \vec{r} = \vec{r}_0 + \frac{m}{qB} v_0 \sin \theta \sin \left(\frac{qB}{m} t\right) (\hat{v}_0 - (\hat{v}_0 \cdot \hat{B}) \hat{B}) \\ + \frac{m}{qB} v_0 \sin \theta \left(\cos \frac{qB}{m} t - 1 \right) (\hat{v}_0 \times \hat{B}) + v_0 \cos \theta t \hat{B}$$

解 b. 1.32.

$$\begin{cases} m\ddot{x} = eB\dot{y} \\ m\ddot{y} = -eB\dot{x} \\ m\ddot{z} = eE \end{cases}$$

$$\begin{aligned} & \vec{r} = \\ & B, z' \\ & \vec{v}_0 \\ & \theta \\ & y \\ & x \end{aligned}$$

解 b. 1.40.

假设: J 与 r 成正比

$$B(r) = \frac{\mu_0}{2\pi r} I \cdot \frac{r^2}{R^2} = \frac{\mu_0 I}{2\pi R^2} r$$

$$J \rightarrow nev = \frac{I}{\pi R^2}$$

$$e v B = e E$$

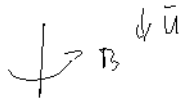
$$\rightarrow E = B v = \frac{\mu_0 I}{2\pi R^2} r \cdot \frac{I}{ne\pi R^2} = \frac{\mu_0 I^2 r}{2\pi^2 R^3 ne}$$

$$\Delta U = \int_0^R E \sigma r$$

$$= \frac{\mu_0 I^2}{16\pi^2 R^3 ne}$$

geg. $\phi = 1, \varphi_0 = \frac{\pi}{2}$

$$B = \frac{\mu_0}{2\pi r} \int_0^r j(r) 2\pi r dr = \frac{1}{r} \int_0^r j(r) r dr$$



$$E = \frac{1}{2\pi \epsilon_0 r} \int_0^r (\rho_0 - \rho_e(r)) 2\pi r dr = \frac{1}{\epsilon_0 r} \int_0^r (\rho_0 - \rho_e(r)) r dr$$

$$j(r) = \rho_e \bar{u}$$

$$e \bar{u} B + e E = 0$$

$$\Rightarrow -\mu_0 \frac{\bar{u}}{r} \int_0^r \rho_e \bar{u} r dr = \frac{1}{\epsilon_0 r} \int_0^r (\rho_0 - \rho_e(r)) r dr$$

$$\hookrightarrow -\bar{u}^2 \int_0^r \rho_e r dr = \frac{1}{\epsilon_0 \mu_0} \int_0^r (\rho_0 - \rho_e) r dr$$

$$\text{mit } \beta = \frac{\bar{u}}{c}$$

$$\Rightarrow -\beta^2 \rho_e = \rho_0 - \rho_e \quad \Rightarrow \quad \rho_e = \frac{1}{1-\beta^2} \rho_0$$